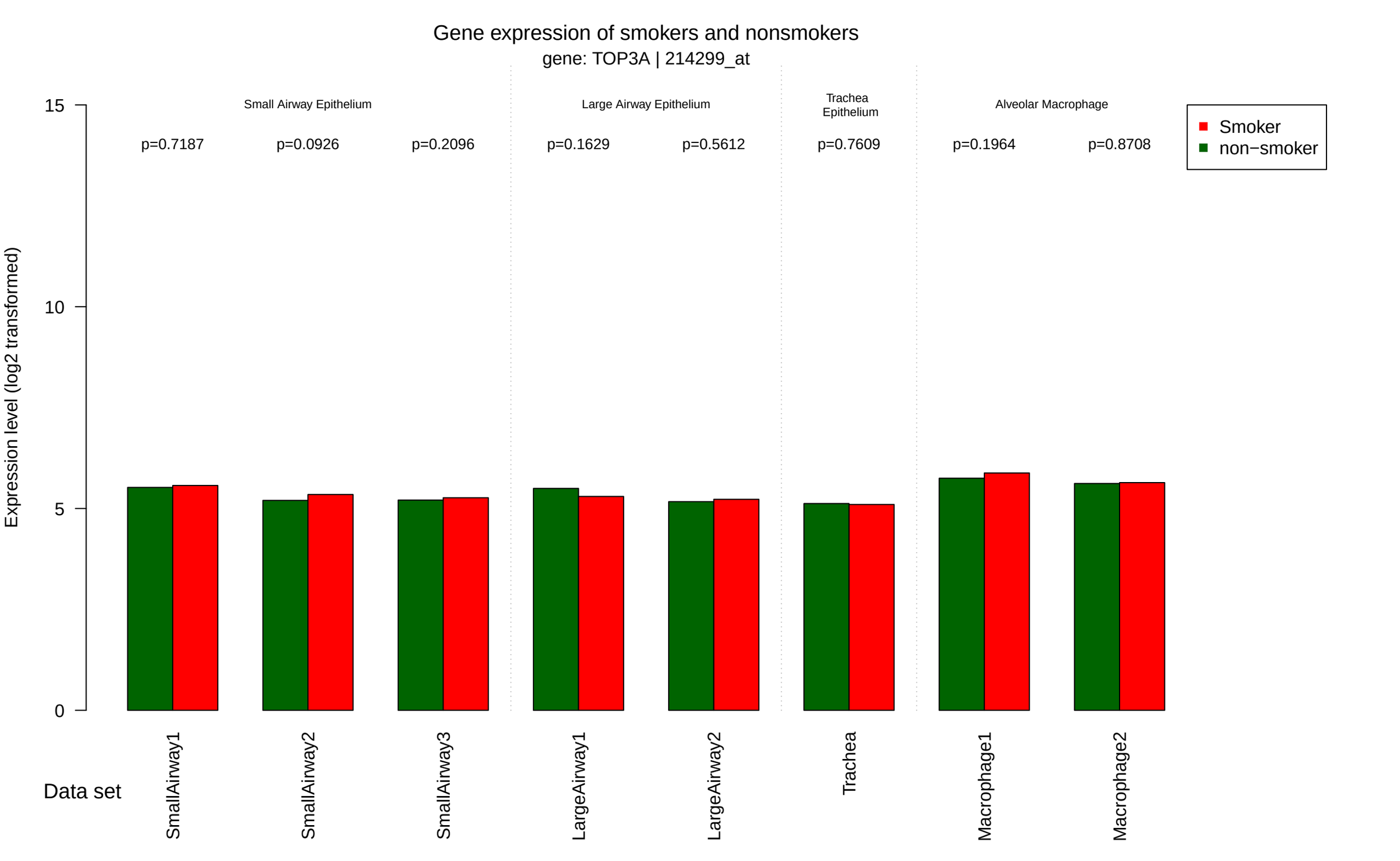
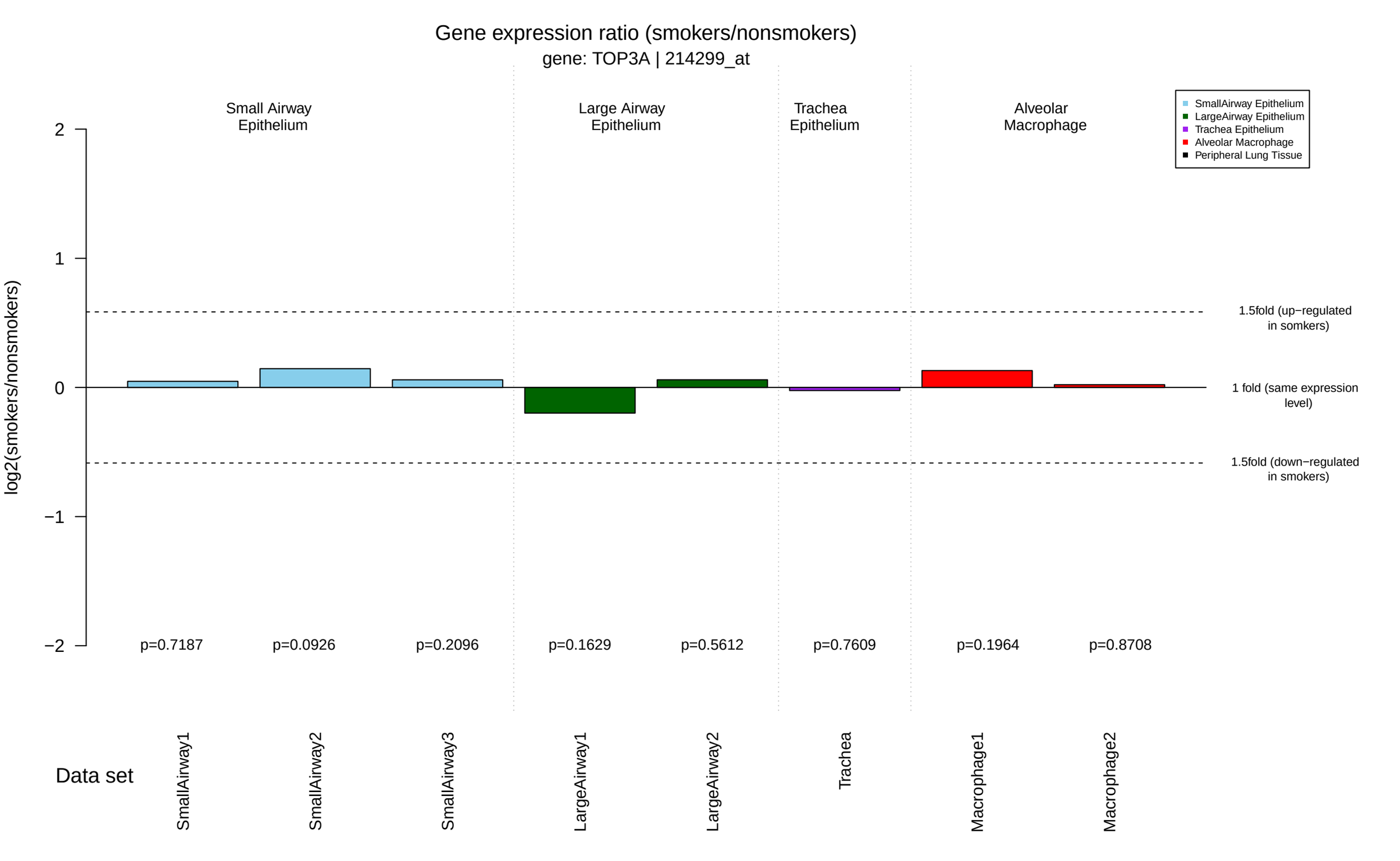
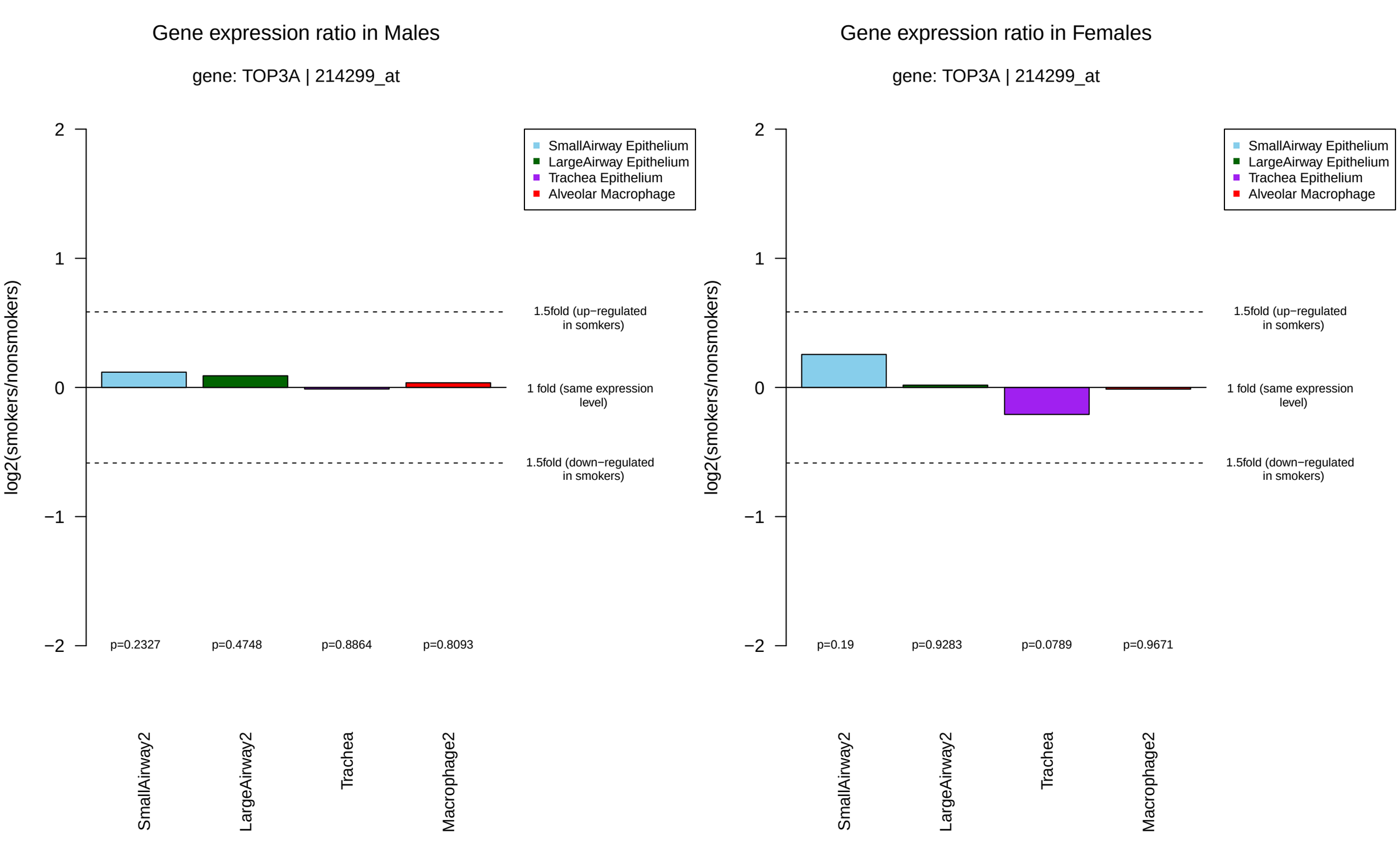
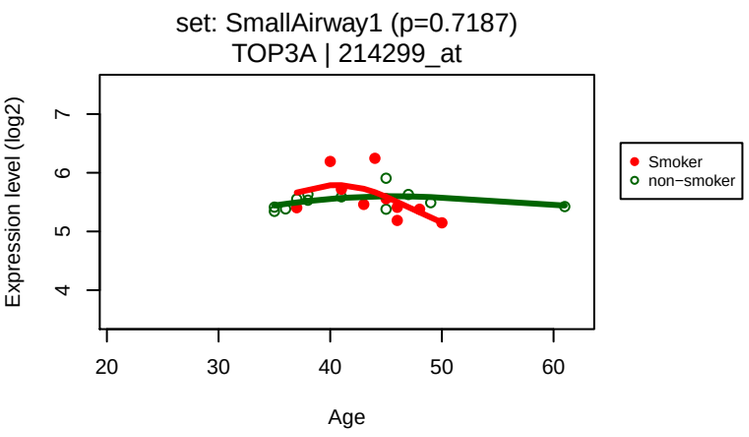
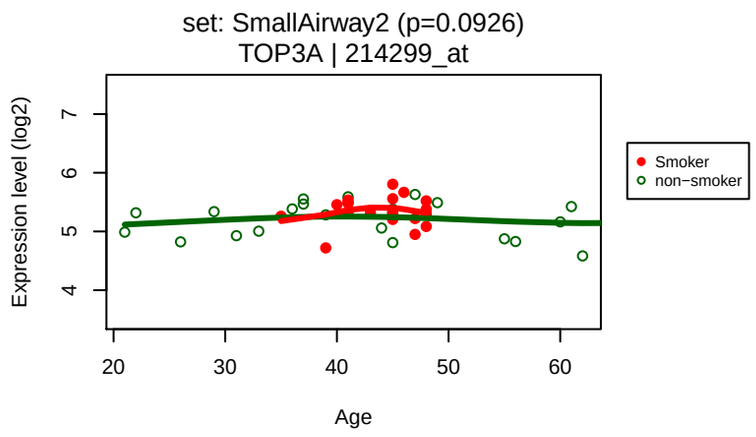
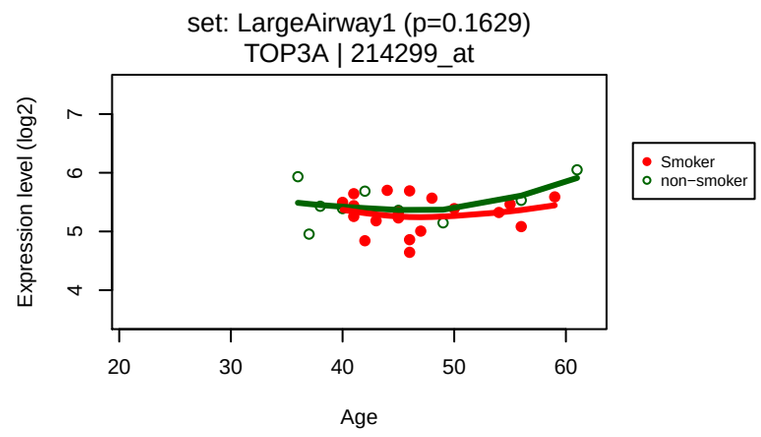

Gene expression of smokers and nonsmokers
gene: TOP3A | 214299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7187
p=0.0926
p=0.2096
p=0.1629
p=0.5612
p=0.7609
p=0.1964
p=0.8708
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

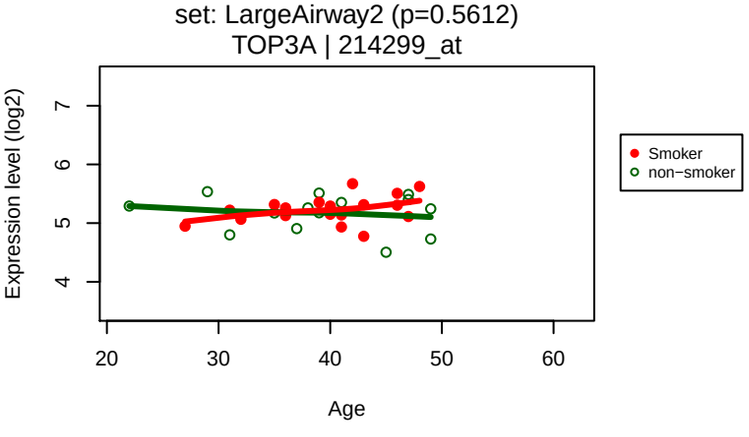

Gene expression ratio (smokers/nonsmokers)
gene: TOP3A | 214299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7187
p=0.0926
p=0.2096
p=0.1629
p=0.5612
p=0.7609
p=0.1964
p=0.8708
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

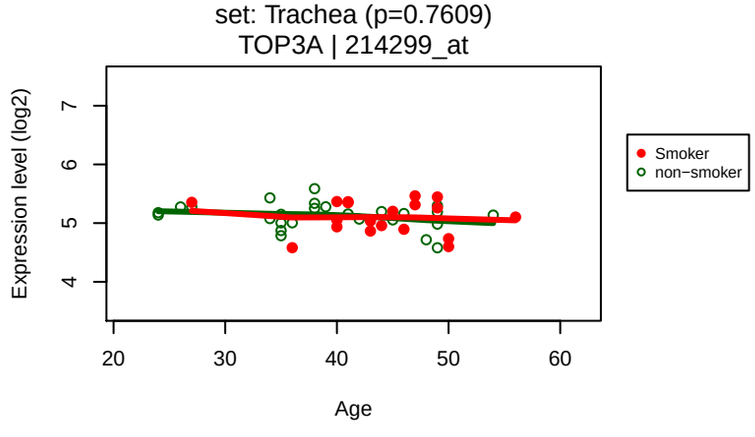

Gene expression ratio in Males
gene: TOP3A | 214299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2327
p=0.4748
p=0.8864
p=0.8093
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP3A | 214299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.19
p=0.9283
p=0.0789
p=0.9671
SmallAirway2
LargeAirway2
Trachea
Macrophage2

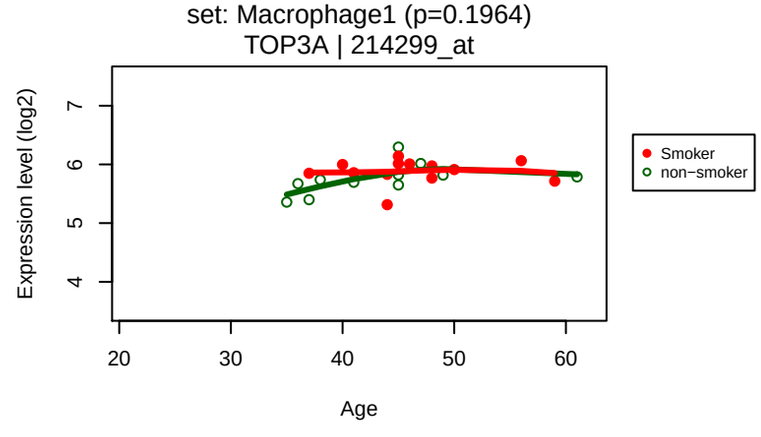

set: SmallAirway1 (p=0.7187)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker

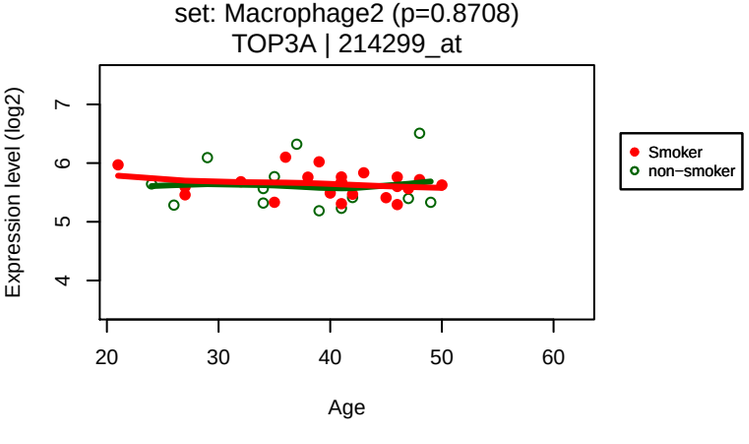

set: SmallAirway2 (p=0.0926)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker

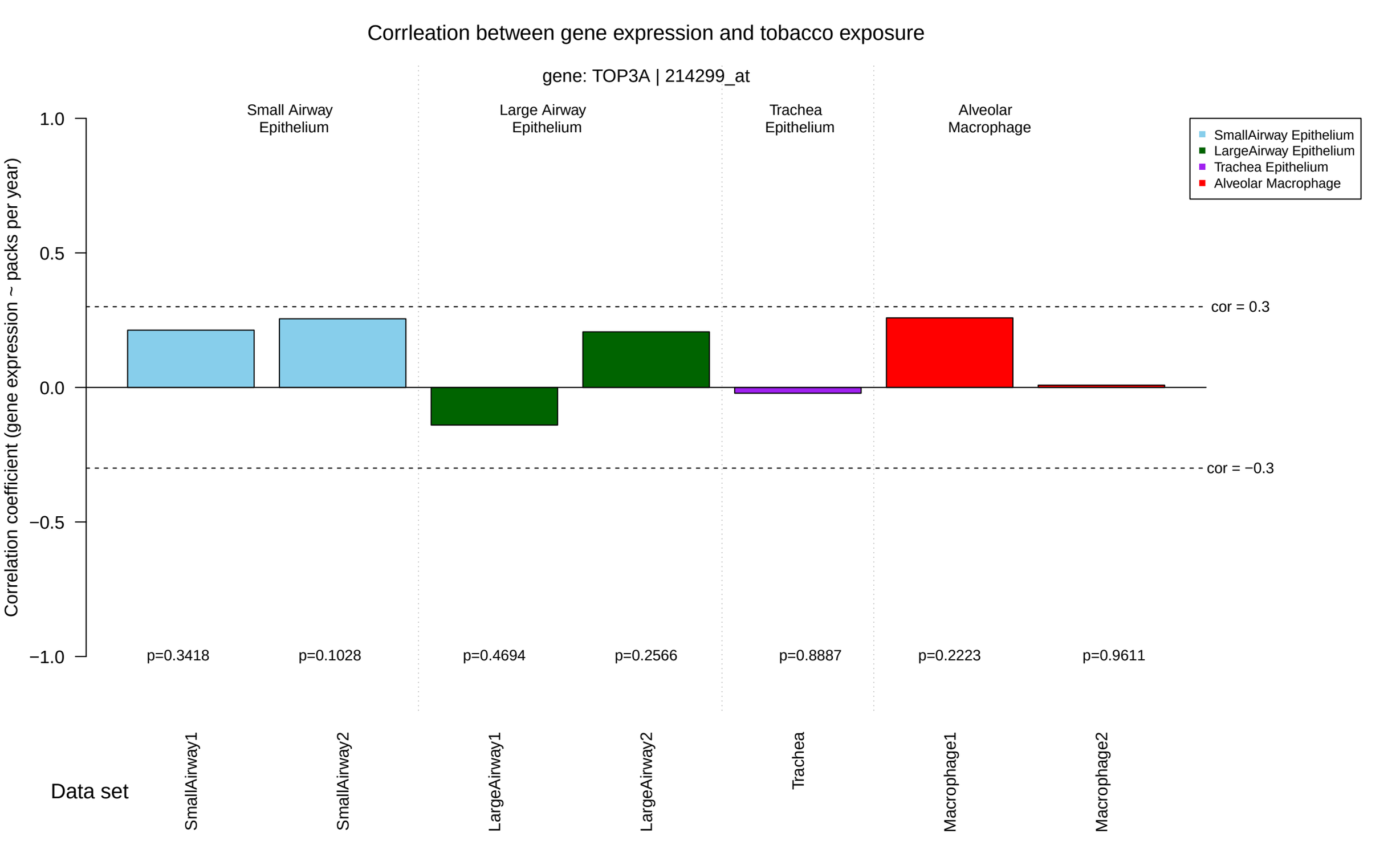

set: LargeAirway1 (p=0.1629)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5612)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7609)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1964)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8708)
TOP3A | 214299_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP3A | 214299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3418
p=0.1028
p=0.4694
p=0.2566
p=0.8887
p=0.2223
p=0.9611
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2